





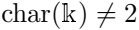
WELSH BOB

WORLD OF
WISDOM

1. INTRODUCTION

Angels and poppies and butterflies.





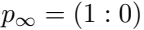
spaced out





Ex 1





Q2 2023

Q E CH 1 X

$$\mathcal{Q}_{d,alg}^2(X) = \{f: \mathbb{P}^1 \rightarrow X \text{ such that } f(\infty) = * \text{ and } f_*([\mathbb{P}^1]) = d\}.$$







$$i: \text{Top}(D^2_{d,\text{alg}}(X)) \rightarrow D^2_{d,\text{top}}(X)$$

Q2, top (X)

0 1 2 3 4 5 6 7



123456











$$i: \mathcal{O}_{d,alg}(X) \rightarrow \mathcal{O}_{d,top}(X)$$

$$i_*: \pi_k(D_{d,alg}^2(X)) \rightarrow \pi_k(D_{d,top}^2(X))$$

W E O X









W E R E

Q2, top (X)



2020

Q200

$$U = \lim_{n \rightarrow \infty} U(n),$$

$$Sp/U = \lim_{n \rightarrow \infty} Sp(2n)/U(n),$$

$$BO = \lim_{n \rightarrow \infty} BO(n),$$

$$(1) \quad \Omega_{d,\mathrm{top}}^2(BU) \simeq BU$$

$$(2) \quad \Omega_{d,\mathrm{top}}^2(Sp/U) \simeq BO$$

$$(3) \quad \Omega_{\mathrm{top}}^2(BO) \simeq O/U$$

$$(4) \quad \Omega_{d,\mathrm{top}}^2(O/U) \simeq BSp$$

$$(5) \quad \Omega_{\mathrm{top}}^2(BSp) \simeq Sp/U$$





12. Sunday, April 1st

Q2 2023



Google is a company that





Instagram





Google AI @ 2019





$$\mathrm{Gr}(n, n+N) = U(n+N)/U(n) \times U(N),$$

$$\mathrm{LGr}(n, 2n) = Sp(2n)/U(n),$$

$$\mathrm{OGr}(n, 2n) = O(2n)/U(n).$$



















$$V = V_{n,d} = \text{Hom}(A, A) \oplus \text{Hom}(V, A) \oplus \text{Hom}(A, U).$$

ERAD





$$v_{d,alg}^2(Gr(n, n+N)) \cong \mathbb{P}^N/G/A).$$

2020-01-01
GIRL, 10
+ 10

$$\sin x + \sin y$$













$$V = V_n, d = \{ (a, j) \in \text{Hom}(A, A) \oplus \text{Hom}(A, U) : a^j \phi = \phi a \}$$

$$\phi(A, \phi) = \{ \phi \in GL(A) : \phi \phi = \phi \}$$



Q&A





$$Q_{d,alg}^2(\mathrm{IGr}(n,2n)) \simeq \mathrm{VO}(A,\phi).$$

022, 2019
LIGI, 2019

$$\ln(x) + \ln(x) + 1$$













$$V = V_n, d = \{ (a, j) \in \text{Hom}(A, A) \oplus \text{Hom}(A, U) : a^j \phi = \phi a \}$$

$$Sp(A, \phi) = \{ \phi \in GL(A) : \phi \phi = \phi \}$$



SpA Sp





$$Q_{d,alg}^2(OGr(n,2n)) \cong \mathbb{R}S/Sp(A,\phi).$$

A pixelated, black and white representation of the text 'V2020-01-01'. The characters are blocky and have a dithered appearance, typical of low-resolution digital art. The text is arranged in a single line.

10-10





A pixelated, black and white representation of the text 'V2020-01-01'. The characters are blocky and have a dithered appearance, typical of low-resolution digital art. The text is arranged in a single line.























13 Topograpia

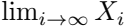
Google + Twitter

Google AI @ 2019

Instagram













$$\begin{aligned}
 & \cdot \cdot \cdot \rightarrow Gr(n, n+N) \rightarrow Gr(n+1, n+N+2) \rightarrow \cdot \cdot \cdot \\
 & \cdot \cdot \cdot \rightarrow Gr(n, n+N) \rightarrow Gr(n+1, n+N+2) \rightarrow \cdot \cdot \cdot
 \end{aligned}$$



$$\lim_{n, N \rightarrow \infty} U(n+N)/(U(n) \times U(N)) = BU.$$

020,216 1-800-234-2345



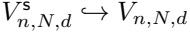
$$\cdot \cdot \cdot \rightarrow \Omega_{\text{alg}}^2(\text{Gr}(n, n+N)) \rightarrow \Omega_{\text{alg}}^2(\text{Gr}(n+1, n+N+2)) \rightarrow \cdot \cdot \cdot$$

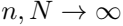
Q2, a16 (BVO)

$$V_n, N_d / GL_d \rightarrow V_{n+1}, N_{d+1} / GL_d \rightarrow \dots$$

VERGIL

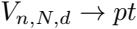
Q2, a1B (BVB)







$$d/dGLd \equiv [d/dGLd]$$



WORLD

IS BIG



$$v_{a, a|g}^2(BV) \approx BV_d.$$

2020-01-01
GIRL, 10
+ 10

BEWE

$$\lim_{n, N \rightarrow \infty} \Omega_{d, \text{alg}}^2(\text{Gr}(n, n+N)) \simeq BV(d).$$







$$\cdot \cdot \cdot \rightarrow \Omega_{a,alg}^2(\mathrm{Gr}(n, n+N)) \rightarrow \Omega_{a+1,alg}^2(\mathrm{Gr}(n+1, n+N+2)) \rightarrow \cdot \cdot \cdot$$

02:00, 20:00

020, 2019 FEBRUARY

Q2 top EV





$$n^2_{\infty, a|b}(BV) \equiv BGL.$$

$$\alpha_{\text{top}}^2(BV) \approx BV.$$

020,216 1-800-451-2263

$$\begin{aligned}
 & \cdot \cdot \cdot \rightarrow Sp(2n)/U(n) \rightarrow Sp(2n+2)/U(n+1) \rightarrow \cdot \cdot \cdot \\
 & \cdot \cdot \cdot \rightarrow Sp(2n)/U(n) \rightarrow Sp(2n+1)/U(n) \rightarrow \cdot \cdot \cdot
 \end{aligned}$$

Q2, a1B, SP, UV





$$\| \sigma^2_{d,alg}(SPV) \| \leq BC(d).$$



$$a^2_{d,alg}(0/0) \simeq BSp(d).$$

$$\lim_{n \rightarrow \infty} \Omega_{d, \text{alg}}^2(\text{Sp}(2n)/U(n)) \simeq BO(d), \quad \lim_{n \rightarrow \infty} \Omega_{d, \text{alg}}^2(O(2n)/U(n)) \simeq B\text{Sp}(d)$$





$$D_{\infty, \text{alg}}^2(\mathcal{S}p/\mathbb{D}) \cong B\mathbb{D}, \quad D_{\infty, \text{alg}}^2(\mathbb{D}/\mathbb{D}) \cong B\mathcal{S}p$$

$$n_{\alpha, \text{top}}^2(Sp/V) \simeq B O, \quad n_{\alpha, \text{top}}^2(O/V) \simeq B Sp$$

23 JAN 1965

23 JAN 1965

23 JAN 1965

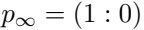
23 JAN 1965

23 JAN 1965

23 JAN 1965

Ex 1







$$\nabla_{\mathbf{K}} \mathcal{L}(\mathbf{V}, \mathbf{K}) = \nabla_{\mathbf{K}} \mathcal{L}(\mathbf{V}, \mathbf{K}) \big|_{\mathbf{K}=\mathbf{K}^*}$$









$$\text{Gr}(n, n+N) = \text{Gr}(n, 0 \oplus N)$$







2020-01-01
GIRL, 10
+ 10

$$0 \longrightarrow F \longrightarrow \underline{U} \oplus \underline{W} \xrightarrow{f} E \longrightarrow 0.$$









1900









$$\underline{V} \xrightarrow{\gamma} E(-p_\infty) \xrightarrow{x_1} E.$$

$$f = \text{def}(x_1, x_2)$$





1900









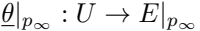


























$$[Hom(A,A) \oplus Hom(A,V) \oplus GL(A)]$$







$(a, j) \in H_m(A, A) \oplus H_m(A, U)$

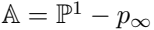
$$\ker(a - \lambda \cdot \text{Id}_A) \cap \ker(v) = 0 \text{ for all } \lambda \in \overline{\mathbb{K}}$$





Ker $\theta = 0$ and Col $\theta = 0$,





$$0 \longrightarrow \underline{U} \xrightarrow{\theta} E \longrightarrow Q \longrightarrow 0$$

$$0 \longrightarrow U \xrightarrow{\theta} H^0(E) \longrightarrow H^0(Q) \longrightarrow 0.$$







$$0 \longrightarrow E(-p_\infty) \xrightarrow{\cdot \underline{x}_1} E \longrightarrow U \otimes \mathcal{O}_{p_\infty} \longrightarrow 0$$

$$0 \longrightarrow H^0(E(-p_\infty)) \xrightarrow{\cdot x_1} H^0(E) \longrightarrow U \longrightarrow 0.$$



ASAP

$$H^0(E) = A \oplus U,$$

$$H^0(E(-1)) = A,$$

$$H^0(Q) = A.$$

$$E \rightarrow E(-1) \xrightarrow{x_0} E$$

$$A \otimes V \xrightarrow{x_0} A \oplus V$$





Oscar A. → J. A. →



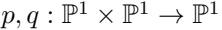




$$0 \longrightarrow \underline{H^0(E(-1))(-1)} \xrightarrow{\underline{x_1x_0-x_0x_1}} \underline{H^0(E)} \longrightarrow E \longrightarrow 0$$

PROBABLY





$$\Delta = \{x_1x_0 - x_0x_1 = 0\} \in \mathbb{P}^1 \times \mathbb{P}^1$$

BEVERLY HILL





$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^1 \times \mathbb{P}^1}(-1, -1) \xrightarrow{y_1 x_0 - y_0 x_1} \mathcal{O}_{\mathbb{P}^1 \times \mathbb{P}^1} \longrightarrow \mathcal{O}_\Delta \longrightarrow 0$$





1990-1990

$$0 \longrightarrow q_*(p^*E(-1, -1)) \longrightarrow q_*(p^*E) \longrightarrow q_*(p^*E) \otimes \mathcal{O}_\Delta \longrightarrow 0$$

$$0 \longrightarrow \underline{H^0(E(-1))(-1)} \xrightarrow{\underline{y_1x_0 - y_0x_1}} \underline{H^0(E)} \longrightarrow E \longrightarrow 0$$

$$x_0, x_1 : \mathbb{R}^0(\mathbb{R}^1) \rightarrow \mathbb{R}^0(\mathbb{R})$$

$$A \xrightarrow{\begin{pmatrix} \alpha \\ j \end{pmatrix}} A \oplus U,$$

$$A \xrightarrow{\begin{pmatrix} \mathrm{Id}_A \\ 0 \end{pmatrix}} A \oplus U$$

$$(7) \quad 0 \longrightarrow \underline{A}(-1) \xrightarrow{\begin{pmatrix} \underline{x}_1 \alpha - \underline{x}_0 \\ \underline{x}_1 j \end{pmatrix}} \underline{A} \oplus \underline{U} \xrightarrow{(\underline{\beta}, \underline{\theta})} E \longrightarrow 0$$





$$\begin{pmatrix} x_1 \alpha - x_0 \\ x_1 \beta \end{pmatrix}$$







$$\begin{pmatrix} x_1 \alpha - x_0 \\ x_1 \beta \end{pmatrix}$$

X

=

20

21

E

101

=

200

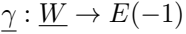
$$\left(a - \lambda \operatorname{Id}_A \right) : A \rightarrow A \oplus U$$

22nd Airborne Division, U.S. Army, Vietnam



$$\underline{U} \oplus \underline{W} \xrightarrow{(\theta, x_1, \gamma)} E.$$







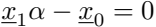












0153





$$(B) \quad \ln(x) + \ln(y) = A \text{ for all } x \in \mathbb{R}.$$













$$V_n, n \in \mathbb{H} \cup \mathbb{A} \oplus \mathbb{H} \cup \mathbb{A} \oplus \mathbb{H} \cup \mathbb{A}$$



QWERTY

Verdun in Verdun



$$\ln(x) = \ln(a) + \ln(b)$$

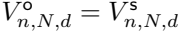


$$v_{d,alg}^2(GI(n, n+N)) \leq v_{n,n,d}/GI(A).$$

GRAND OPENING

ANALYSIS OF ANOVA

POPE JOHN



W E I A R E

equal to zero

































3. PROPERTIES AND 2

$$OG(n, 2n) = O(2n) / O(n), \quad IG(n, 2n) = Sp(2n) / O(n)$$







$$X_n = O(2n)/O(n), \quad X_n = Sp(2n)/U(n).$$



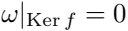






















$$0 \longrightarrow I^v \xrightarrow{j^{-1} \circ f^v} V \xrightarrow{f} I \longrightarrow 0$$





$$j_{\pm} = \begin{pmatrix} 0 & \pm 1 v v \\ 1 v & 0 \end{pmatrix}.$$







$$0 \longrightarrow L^V \xrightarrow{j_{\pm}^{-1} \circ f^V} U \oplus U^V \xrightarrow{f} L \longrightarrow 0.$$



[illegible]

02da18X±m



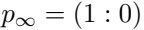
$$(9) \quad 0 \longrightarrow E^V \xrightarrow{j_{\pm}^{-1} \circ f^V} \underline{U} \oplus \underline{U}^V \xrightarrow{f} E \longrightarrow 0$$







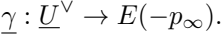






$$f = \frac{1}{2\pi} \frac{d}{dt} \ln \left(\frac{1}{\sqrt{1 - \beta^2}} \right)$$





$$j_{\pm}^{-1} \circ f^v = \begin{pmatrix} 2^v x_1 \\ \pm \theta^v \end{pmatrix}$$

for love

$$(10) \quad \vartheta_1 \vartheta_1^v = F \vartheta_1 \vartheta_1^v.$$

Q2, a, b, X, ±, ∓

100%



(11)

$$\phi^\vee = \mp \phi$$

$$\phi \alpha = \alpha^\vee \phi$$

$$\phi \gamma = -j^\vee.$$



ERAD

WALLS



ALPHAS



$$0 \rightarrow E^{\vee}(-1) \rightarrow \bigoplus E^{\vee}(-1) \rightarrow E(-1) \rightarrow 0.$$

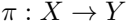


ANDERSON



$$\phi: A \rightarrow A^0(E(-1)) \rightarrow A^1(E^{\vee}(-1)) \rightarrow A^0(E(-1))^{\vee} = A^{\vee}.$$





WORLDWIDE

$$\mathbb{D}_X(F) = (F)^{\vee} \otimes_X [\mathrm{d}i_X]$$







$$F^{\bullet} = [U(-1) \xrightarrow{\theta} E(-1)]$$





$$\mathbb{D}_1 F^\bullet = [E^V(-1) \xrightarrow{\theta^V} U^V(-1)].$$





$$0 \longrightarrow E^\vee \xrightarrow{\begin{pmatrix} \underline{\gamma}^\vee & \underline{x}_1 \\ \pm \underline{\theta}^\vee \end{pmatrix}} \underline{U} \oplus \underline{U}^\vee \xrightarrow{(\underline{\theta}, \underline{x}_1 \underline{\gamma})} E \longrightarrow 0$$

$$\begin{array}{ccc}
 \underline{U}(-1) & \xrightarrow{\underline{\theta}} & E(-1) \\
 \uparrow \mp \underline{\gamma}^\vee \underline{x}_1 & & \uparrow \underline{x}_1 \underline{\gamma} \\
 E^\vee(-1) & \xrightarrow{\underline{\theta}^\vee} & \underline{U}^\vee(-1)
 \end{array}$$

$$\begin{array}{ccc}
 F^\bullet & & \\
 \uparrow \Psi & \curvearrowright \Phi = \Psi^{-1} & \\
 \mathbb{D}F^\bullet & &
 \end{array}$$

















$$\begin{array}{ccc}
 \underline{U}^{\vee}(-1) & \xleftarrow{\underline{\theta}^{\vee}} & E^{\vee}(-1) \\
 \downarrow \mp \underline{x}_1 \underline{\gamma} & & \downarrow \underline{\gamma}^{\vee} \underline{x}_1 \\
 E(-1) & \xleftarrow{\underline{\theta}} & \underline{U}(-1)
 \end{array}$$

$$\begin{array}{ccc}
 \mathbb{D}F^{\bullet} & & \\
 \downarrow \mathbb{D}\Psi & & \\
 F^{\bullet} & &
 \end{array}$$









$$R\pi_*\mathcal{O} = R\pi_*\mathcal{F}^{\bullet} = A \rightarrow R\pi_*\mathcal{D}_p\mathcal{F}^{\bullet} = \mathcal{D}_p R\pi_*\mathcal{F}^{\bullet} = A'$$



$$\begin{aligned}
\phi &= R\pi_*\bar{\Phi} \\
&= \mp R\pi_*\mathbb{D}_{\mathbb{P}^1}\bar{\Phi} \\
&= \mp \mathbb{D}_{pt}R\pi_*\bar{\Phi} \\
&= \mp \mathbb{D}_{pt}\phi \\
&= \mp \phi^\vee
\end{aligned}$$

[illegible]



$$0 \longrightarrow \underline{A}(-2) \xrightarrow{\begin{pmatrix} \underline{\sigma} \\ \underline{x_1j} \end{pmatrix}} (\underline{A} \oplus \underline{U})(-1) \xrightarrow{(\underline{\beta}, \underline{\theta})} E(-1) \longrightarrow 0.$$

$$\begin{array}{ccc}
 \underline{A}(-2) & \xrightarrow{\underline{\sigma}} & \underline{A}(-1) \\
 \downarrow -\underline{x}_1 j & & \downarrow \underline{\beta} \\
 \underline{U}(-1) & \xrightarrow{\underline{\theta}} & E(-1)
 \end{array}$$

$$\begin{array}{c}
 A^\bullet \\
 \downarrow \mathbb{J} \\
 F^\bullet
 \end{array}$$









(12)

$$\begin{array}{ccc}
\underline{A}(-2) & \xrightarrow{\underline{\sigma}} & \underline{A}(-1) \\
\downarrow -\underline{x}_1 j & & \downarrow \underline{\beta} \\
\underline{U}(-1) & \xrightarrow{\underline{\theta}} & E(-1) \\
\uparrow \mp \underline{\gamma}^\vee \underline{x}_1 & & \uparrow \underline{x}_1 \underline{\gamma} \\
E^\vee(-1) & \xrightarrow{\underline{\theta}^\vee} & \underline{U}^\vee(-1) \\
\downarrow \underline{\beta}^\vee & & \downarrow -\underline{x}_1 j^\vee \\
\underline{A}^\vee(-1) & \xrightarrow{\underline{\sigma}^\vee} & \underline{A}^\vee
\end{array}$$

$$\begin{array}{c}
A^\bullet \\
\downarrow \mathbb{J} \\
F^\bullet \\
\downarrow \Phi \\
\mathbb{D}F^\bullet \\
\downarrow \mathbb{D}\mathbb{J} \\
\mathbb{D}A^\bullet
\end{array}$$



11

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10

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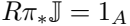


APR 20 1964













SPENCER
WILSON

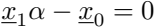












Google



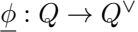














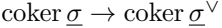






$$\begin{array}{ccc}
 \underline{A}^{\vee} & \xrightarrow{\underline{\sigma}^{\vee}} & \underline{A}^{\vee}(1) \\
 \uparrow \phi \underline{x}_1 & & \uparrow \phi \underline{x}_1 \\
 \underline{A}(-1) & \xrightarrow{\underline{\sigma}} & \underline{A}
 \end{array}$$

$$\begin{array}{c}
 \mathbb{D}A^{\bullet} \\
 \uparrow \underline{\phi} \\
 A^{\bullet}
 \end{array}$$









$$\begin{array}{ccccc}
\underline{A}(-1) & \xrightarrow{\underline{\sigma}} & \underline{A} & \longrightarrow & \operatorname{coker} \underline{\sigma} \\
\downarrow -\underline{x}_1 j & & \downarrow \underline{\beta} & & \downarrow \\
\underline{U} & \xrightarrow{\underline{\theta}} & \underline{E} & \longrightarrow & \operatorname{coker} \underline{\theta} \\
\uparrow \mp \underline{\gamma}^\vee \underline{x}_1 & & \uparrow \underline{x}_1 \underline{\gamma} & & \uparrow \\
\underline{E}^\vee & \xrightarrow{\underline{\theta}^\vee} & \underline{U}^\vee & \longrightarrow & \operatorname{coker} \underline{\theta}^\vee \\
\downarrow \underline{\beta}^\vee & & \downarrow -\underline{x}_1 j^\vee & & \downarrow \\
\underline{A}^\vee & \xrightarrow{\underline{\sigma}^\vee} & \underline{A}^\vee(1) & \longrightarrow & \operatorname{coker} \underline{\sigma}^\vee \\
\uparrow \phi \underline{x}_1 & & \uparrow \phi \underline{x}_1 & & \uparrow \\
\underline{A}(-1) & \xrightarrow{\underline{\sigma}} & \underline{A} & \longrightarrow & \operatorname{coker} \underline{\sigma}
\end{array}$$

$$\begin{array}{ccccccc}
0 & \longrightarrow & A & \xrightarrow{1_A} & A & \longrightarrow & 0 \\
\downarrow & & \downarrow \begin{pmatrix} 1_A \\ 0 \end{pmatrix} & (1) & \downarrow 1_A & & \downarrow \\
0 & \longrightarrow & U & \xrightarrow{\begin{pmatrix} 0 \\ 1_U \end{pmatrix}} & A \oplus U & \xrightarrow{B_1} & A \longrightarrow 0 \\
\uparrow & & \uparrow \begin{pmatrix} \gamma \\ 0 \end{pmatrix} & (2) & \uparrow \phi^{-1} & & \uparrow \\
0 & \longrightarrow & H^0(E^\vee) & \longrightarrow & U^\vee & \xrightarrow{B_2} & A^\vee \longrightarrow H^1(E^\vee) \\
\downarrow & & \downarrow \begin{pmatrix} 0 \\ -j^\vee \end{pmatrix} & (3) & \downarrow 1_{A^\vee} & & \downarrow \\
0 & \longrightarrow & A^\vee & \xrightarrow{\begin{pmatrix} -1_{A^\vee} \\ \alpha^\vee \end{pmatrix}} & A^\vee \oplus A^\vee & \xrightarrow{B_3} & A^\vee \longrightarrow 0 \\
\uparrow & & \uparrow \begin{pmatrix} 0 \\ \phi \end{pmatrix} & (4) & \uparrow \phi & & \uparrow \\
0 & \longrightarrow & A & \xrightarrow{1_A} & A & \longrightarrow & 0
\end{array}$$



1A20



1011101





$$v_{d,alg}^2(x_{\pm},n) = v_{F,n,d}/G(A)$$



$$(a, \gamma, j, \phi) \in \text{Hom}(A, A) \oplus \text{Hom}(E, A) \oplus \text{Hom}(A, E) \oplus \text{Hom}(A, A)$$

WFO
for
G

WFO
for
d

Verdun in Verdun



$\ln(x) \cdot \ln(x) + \ln(x) = \ln(x)^2 + \ln(x)$





ERAD

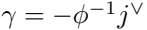




$$\text{GL}(A, \phi) = \{ \phi \in \text{GL}(A) : \phi \phi = \phi \}.$$







$$D_{d,alg}^2(X_{\pm}, n) = \mathrm{Vol}_{F,n}(\mathrm{Gr}(A, \phi))$$

As a







$$V_{F,n,\alpha} = \{ (a, j) \in \text{Hom}(A, A) \oplus \text{Hom}(A, V) : \phi a = a \phi \},$$



Verdun in Verdun



$\ln(x) = \ln(x) + \ln(1) = \ln(x)$



$$\operatorname{Im}(\alpha - \lambda \cdot \operatorname{Id}_A) + \operatorname{Im}(\phi^{-1}j^\vee) = A \text{ for all } \lambda \iff$$

$$\operatorname{Im}(\phi\alpha\phi^{-1} - \lambda \cdot \operatorname{Id}_{A^\vee}) + \operatorname{Im}(j^\vee) = A^\vee \text{ for all } \lambda \iff$$

$$\operatorname{Im}(\alpha^\vee - \lambda \cdot \operatorname{Id}_{A^\vee}) + \operatorname{Im}(j^\vee) = A^\vee \text{ for all } \lambda \iff$$

$$\operatorname{Ker}(\alpha - \lambda \cdot \operatorname{Id}_A) \cap \operatorname{Ker}(j) = 0 \text{ for all } \lambda.$$









As a









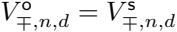
$$V_{\mp, n, d} = \{ (a, j) \in \text{Hom}(A, A) \oplus \text{Hom}(A, B) : \phi a = a^{\vee} \phi \}$$



$$\text{Ker}(a - \lambda \cdot 1_A) \cap \text{Ker}(a) = 0 \text{ for all } \lambda \in \overline{\mathbb{K}}.$$

$$D_{d, \text{align}}^2(X_{\pm}, n) = \text{Vol}_{F, n}(\text{GRA}, \emptyset)$$

GA, 1994
GA, 1994



ES

F

1

2

3

4

Google



LEVEL OF
SOCIETY

LEVEL OF
SOCIETY
IN THE
COUNTRY



$$E^V \xrightarrow{j_{\pm}^{-1} \circ f^V} \underline{U} \oplus \underline{U}^V \xrightarrow{f} E,$$















Google

Q. I. E. V. P. S.

Google

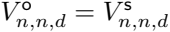
A pixelated, black and white graphic of the text "G.A.P. is a good idea". The text is rendered in a stylized, hand-drawn font where each letter is composed of individual pixels, giving it a retro, digital appearance. The letters are thick and blocky, with some internal pixel detail visible. The spacing between the letters is consistent, and the overall composition is centered horizontally. The background is plain white, making the black and gray pixels of the text stand out.





QWERTYUIOPASDFGHJKLZXCVBNM

Q. A. 15. V. E. V. n. d.



ERAD

ERAD



WAGGERS

[illegible]



A pixelated, black and white graphic of the text "G.A.P. is a good idea". The text is rendered in a stylized, hand-drawn font where each letter is composed of individual pixels, giving it a retro, digital appearance. The letters are bold and slightly irregular, with varying shades of gray and black pixels. The word "G.A.P." is followed by a period, and the phrase "is a good idea" follows in a similar style. The entire graphic is set against a plain white background.

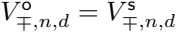






Google





4. IDENTIFY THE TOPICAL AREA



AdGc $\xrightarrow{\text{top}}$ AdTop

Handwritten text: "HUGO" in a stylized, cursive script.

Handwritten text: "GO" in a stylized, cursive script.

Handwritten text: "HUGO" in a stylized, cursive script.





4062

$$\cdot \cdot \cdot \rightarrow \Omega_{\text{alg}}^2(\text{Gr}(n, n+N)) \rightarrow \Omega_{\text{alg}}^2(\text{Gr}(n+1, n+N+2)) \rightarrow \cdot \cdot \cdot$$



Q2, a16 (BVO)

$$a^2_{\alpha, \alpha|B}(BV) \equiv BG|_d$$



2020-01-01
GIRL, 10
+ 10

BEWE

$$\lim_{n, N \rightarrow \infty} \Omega_{d, \text{alg}}^2(\text{Gr}(n, n+N)) \simeq BV(d).$$

$$\cdot \cdot \cdot \rightarrow \Omega_{a,alg}^2(\mathrm{Gr}(n, n+N)) \rightarrow \Omega_{a+1,alg}^2(\mathrm{Gr}(n+1, n+N+2)) \rightarrow \cdot \cdot \cdot$$



020, 2019 FEBRUARY

$$n^2_{\alpha, \beta}(BV) \simeq BG$$





$$\lim_{n, N \rightarrow \infty} \text{Gr}(n, n+N) \simeq BU,$$

$$d, n, N \rightarrow \infty \quad \Omega_{d, \text{alg}}^2(\text{Gr}(n, n+N)) \simeq \Omega_{d, \text{top}}^2(BU),$$



$$\alpha_{\text{top}}^2(BV) \approx BV.$$

$$X_n = O(2n)/O(n) = OI(n, 2n) \quad X_n = Sp(2n)/O(n) = LO(n, 2n).$$

$$\begin{aligned}
 & \cdot \cdot \cdot \rightarrow d_{\text{alg}}^2(X_{\pm}, n) \rightarrow d_{\text{alg}}^2(X_{\pm}, n+1) \rightarrow \cdot \cdot \cdot \\
 & \cdot \cdot \cdot \rightarrow d_{\text{alg}}^2(X_{\pm}, n) \rightarrow d_{\text{alg}}^2(X_{\pm}, n+1) \rightarrow \cdot \cdot \cdot
 \end{aligned}$$



$$\Omega^2_{d,\mathrm{alg}}(X_+,\infty)\cong BSp(d)$$

$$\Omega^2_{d,\mathrm{alg}}(X_-,\infty)\cong BO(d)$$



02da18X±m

$$\lim_{n \rightarrow \infty} \Omega^2_{d,\mathrm{alg}}(X_+, n) \simeq BSp(d)$$

$$\lim_{n \rightarrow \infty} \Omega^2_{d,\mathrm{alg}}(X_-, n) \simeq BO(d).$$

$$\cdot \cdot \cdot \rightarrow d_{\text{alg}}^2(X_{\pm}, n) \rightarrow d_{d+2, \text{alg}}^2(X_{\pm}, n+2) \rightarrow \cdot \cdot \cdot$$



1234567890

$$\Omega^2_{\infty,\mathrm{alg}}(X_+,\infty)\cong BSp$$

$$\Omega^2_{\infty,\mathrm{alg}}(X_-,\infty)\cong BO$$







$$\lim_{n \rightarrow \infty} X_{+,n} = 0/U, \quad \lim_{n \rightarrow \infty} X_{-,n} = Sp/U,$$

$$\lim_{d,n\rightarrow\infty}\Omega^2_{d,\mathrm{alg}}(X_{+,n})\simeq\Omega^2_{d,\mathrm{top}}(O/U),$$

$$\lim_{d,n\rightarrow\infty}\Omega^2_{d,\mathrm{alg}}(X_{-,n})\simeq\Omega^2_{d,\mathrm{top}}(Sp/U)$$



$$\Omega^2_{d,\mathrm{top}}(O/U) \simeq BSp,$$

$$\Omega^2_{d,\mathrm{top}}(Sp/U) \simeq BO.$$

41. *Impatiens* *capensis*

020,216 1-800-451-2263

$$Gr(n, n+N) \rightarrow Gr(n+1, n+N+2).$$



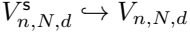
$$\left\{ \underline{U} \oplus \underline{W} \xrightarrow{(f_v, f_w)} E \right\} \mapsto \left\{ \underline{U} \oplus \mathcal{O} \oplus \underline{W} \oplus \mathcal{O} \xrightarrow{(f_v \oplus \text{Id}, f_w \oplus 0)} E \oplus \mathcal{O} \right\}$$

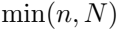
$$(A, D, M, \alpha, \gamma) \mapsto (A, D \oplus K, M \oplus K, \alpha, \gamma, 0), (n, 0)$$

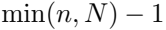
(13)

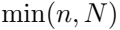
$$\begin{array}{ccccccc}
\cdots & \hookrightarrow & V_{n,N,d}^s/GL_d & \hookrightarrow & V_{n+1,N+1,d}^s/GL_d & \hookrightarrow & \cdots \\
& & \downarrow & & \downarrow & & \\
\cdots & \hookrightarrow & [V_{n,N,d}/GL_d] & \hookrightarrow & [V_{n+1,N+1,d}/GL_d] & \hookrightarrow & \cdots
\end{array}$$











1. The first part of the document discusses the importance of maintaining accurate records of all transactions. It emphasizes that proper record-keeping is essential for ensuring the integrity and transparency of financial data. This section also outlines the various methods used to collect and analyze data, highlighting the role of technology in streamlining these processes.

2. The second part of the document focuses on the challenges faced by organizations in implementing effective data management strategies. It explores the complexities of integrating disparate data sources and the need for robust security measures to protect sensitive information. Additionally, it discusses the importance of training staff to ensure they are equipped with the necessary skills to handle data responsibly.

3. The third part of the document provides a detailed overview of the current state of data management practices across different industries. It compares various approaches and identifies best practices that can be adopted to improve efficiency and accuracy. This section also addresses the evolving regulatory landscape and the impact of emerging technologies on data management.

4. The final part of the document offers practical recommendations for organizations looking to optimize their data management processes. It suggests specific steps for assessing current capabilities, identifying areas for improvement, and implementing targeted solutions. The document concludes by emphasizing the ongoing nature of data management and the need for continuous monitoring and adaptation to changing requirements.

$$\frac{d}{dt} \left(\frac{1}{2} m v^2 + U(r) \right) = 0$$



→ U → V → W → X → Y → Z

$$\dim(V_{n,d}^S/GI_d) = \dim[V_{n,d}/GI_d] = (n+N)d.$$

$$\{ \underline{U} \oplus \underline{W} \rightarrow E \}$$









code is for

2021-2022

VS. W.D. 1/10/2011

$$\left\{ \underline{U} \oplus \underline{W} \xrightarrow{f} E \right\} \mapsto \left\{ \underline{U} \oplus \underline{W} \xrightarrow{f'} E' \right\}$$





$\rightarrow E \Rightarrow E \Rightarrow O \Rightarrow E \Rightarrow E \Rightarrow O \Rightarrow E \Rightarrow O.$

$$\dim(P(\text{Ext}^1(\mathcal{O}_p, E))) + \dim(P^1 - p_\infty) = (n-1) + 1 = n.$$

$$x + d - 1/x + d$$

1. The first part of the document discusses the importance of maintaining accurate records of all transactions. It emphasizes that proper record-keeping is essential for ensuring the integrity and transparency of financial data. This section also outlines the various methods used to collect and analyze data, highlighting the role of technology in streamlining these processes.

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3. The third part of the document provides a detailed overview of the current state of data management practices across different industries. It compares various approaches and identifies best practices that can be adopted to improve efficiency and accuracy. This section also includes case studies that demonstrate the successful implementation of data management solutions in real-world scenarios.

4. The fourth part of the document discusses the future of data management, highlighting emerging trends and technologies that will shape the landscape. It explores the potential of artificial intelligence and machine learning in automating data analysis and decision-making processes. Furthermore, it addresses the ethical considerations surrounding data usage and the need for regulatory frameworks to govern the collection and processing of personal data.

$$d(N+x) - x - (d-1)(N+x) = N.$$



Q2, a1B (BVB)



2020-2021
BIOLOGY



Handwritten text: $[v, v, v, v] \rightarrow Bv$

42. Improving the world

$$\Omega_{d,alg}^2(\mathrm{Gr}(n, n+N)) \rightarrow \Omega_{d+1,alg}^2(\mathrm{Gr}(n+1, n+N+2))$$

$$\left\{ \underline{U} \oplus \underline{W} \xrightarrow{(f_U, f_W)} E \right\} \mapsto \left\{ \underline{U} \oplus \mathcal{O} \oplus \underline{W} \oplus \mathcal{O} \xrightarrow{\begin{pmatrix} f_U & 0 & f_W & 0 \\ 0 & \underline{x}_0 & 0 & \underline{x}_1 \end{pmatrix}} E \oplus \mathcal{O}(1) \right\}$$

$$(A, U, M, (a, s, n)) \mapsto (A, U, M, (a, s, n)) \oplus (B, K, K, (0, 1, 1)).$$

$$\cdots \hookrightarrow V_{n,N,d}^s/GL_d \hookrightarrow V_{n+1,N+1,d+1}^s/GL_{d+1} \hookrightarrow \cdots$$

$$\downarrow$$

$$\downarrow$$

$$\cdots \longrightarrow [V_{n,N,d}/GL_d] \longrightarrow [V_{n+1,N+1,d+1}/GL_{d+1}] \longrightarrow \cdots$$



1. *W. G. L.* *W. G. L.*



020, 2019
EVE

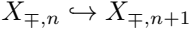




BCI → BCI_{d+1} → .

43. **Impressions**

020,216 1-800-234-2345





$$\left\{ \underline{U} \oplus \underline{U}^v \xrightarrow{(f_U, f_{U^v})} E \right\} \mapsto \left\{ \underline{U} \oplus 0 \oplus \underline{U}^v \oplus 0 \xrightarrow{(f_U, \text{Id}, f_{U^v}, 0)} E \oplus 0 \right\}.$$

$$\{A, \phi, U, (a, j)\} \mapsto \{A, \phi, U \oplus \mathbb{R}, (a, j, 0)\}.$$

(14)

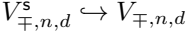
$$\begin{array}{ccccccc}
 \cdots & \hookrightarrow & V_{\mp,n,d}^s/G(A,\phi) & \hookrightarrow & V_{\mp,n+1,d}^s/G(A,\phi) & \hookrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \\
 \cdots & \hookrightarrow & [V_{\mp,n,d}/G(A,\phi)] & \hookrightarrow & [V_{\mp,n+1,d}/G(A,\phi)] & \hookrightarrow & \cdots
 \end{array}$$

















$$Z = \{(a, j) \in V_{F, n, d} : \exists \lambda \in \overline{\mathbb{K}}, v \in A \text{ such that } j(v) = 0 \text{ and } av = \lambda v\}.$$





$$V_+ = \mathrm{Sym}^2 A^\vee \oplus \mathrm{Hom}(A, U),$$

$$V_- = \Lambda^2 A^\vee \oplus \mathrm{Hom}(A, U).$$

we are all



$$B(v, w) = (v, w)$$









$$\sqrt{v, w} = \sqrt{w, v}$$







$$B_0(v) = A(v)$$

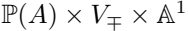


$$Z = \{ (B_a, j) \in V_+ : \exists \lambda \in \overline{\mathbb{K}}, v \in A \text{ such that } j(v) = 0 \text{ and } B_a(-, v) = \lambda(-, v) \}.$$



$$\tilde{Z} = \{([v], (B_\alpha, j), \lambda) \in \mathbb{P}(A) \times V_+ \times A^1 : jv = 0, B_\alpha(-, v) = \lambda(-, v)\}.$$

2019

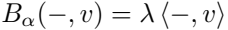














$$v \in \mathbb{R}^n : A^T v \in \mathcal{C} \implies A^T v \in \mathcal{C}^*$$

$$B_d(u, \lambda) \mapsto B_d(u, v) - \lambda(u, v).$$



WIZARD







$$\mathrm{Sym}^2 A^\vee \oplus A^1 \xrightarrow{L_v} A^\vee$$

$$\Lambda^2 A^\vee \oplus A^1 \xrightarrow{L_v} A^\vee.$$

$$A^v \xrightarrow{ev_v} A^1$$

$$f(-) \mapsto f(v)$$

$$B_{\alpha}(\psi, \psi) = \psi, \psi) = 0$$

$$L(B, X) = 0$$























$$A \times V \rightarrow V \quad \text{PAA} \times V \rightarrow V$$



$$\mathrm{codim}(\tilde{Z}) - d = \begin{cases} n & \text{for } V_+ \\ n - 1 & \text{for } V_- \end{cases}$$



GAPOPOPO



GOVERNMENT



BEAWE



4.4. Example

$$D_{d,d|g}^2(X_{\pm},n) \rightarrow D_{d+2,d|g}^2(X_{\pm},n+2)$$

$$\{(A, \phi), \tau, (a, j)\} \mapsto \{(A \oplus A^2, \phi \oplus \phi_0), \tau \oplus A^2, (a \oplus a_0, j \oplus j_0)\}$$

$$\phi_0 = \begin{pmatrix} 0 & 1 \\ \mp 1 & 0 \end{pmatrix} \quad a_0 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad j_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

$$\left\{ \underline{U} \oplus \underline{U}^\vee \xrightarrow{(f_U, f_{U^\vee})} E \right\} \mapsto \left\{ \underline{U} \oplus \mathcal{O}^2 \oplus \underline{U}^\vee \oplus \mathcal{O}^2 \xrightarrow{\begin{pmatrix} f_U & 0 & 0 & f_{U^\vee} & 0 & 0 \\ 0 & x_0 & 0 & 0 & 0 & \pm x_1 \\ 0 & 0 & x_0 & 0 & -x_1 & 0 \end{pmatrix}} E \oplus \mathcal{O}(1) \right\}$$

(15)

$$\begin{array}{ccccccc}
 \cdots & \hookrightarrow & V_{\mp,n,d}^s/G_{\mp}(d) & \longrightarrow & V_{\mp,n+2,d+2}^s/G_{\mp}(d+2) & \hookrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \\
 \cdots & \longrightarrow & [V_{\mp,n,d}/G_{\mp}(d)] & \longrightarrow & [V_{\mp,n+2,d+2}/G_{\mp}(d+2)] & \longrightarrow & \cdots
 \end{array}$$

Google









BEFORE + AFTER

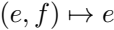






$$Y_2 = \{ (e, f) \in A^2 : (e, f) = 1 \}.$$

1234567

















Google

$$BC(d) \equiv [V(d) + 2].$$

$$\begin{aligned}
BG_-(d) &\xrightarrow{\cong} [Y_-/G_-(d+2)] \\
&\xrightarrow{\text{affine bundle}} [A - \{0\}/G_-(d+2)] \\
&\xrightarrow{(d+1)\text{-connected}} [A/G_-(d+2)] \\
&\xrightarrow{\text{affine bundle}} BG_-(d+2).
\end{aligned}$$

BEFORE + AFTER









$$Y_4 = \{(e, f) \in A^2 : (e, e) = 0, (f, f) = 1\}.$$













$$= [v + g + d + 2].$$

$$[y+g+d+2] \rightarrow B+g+d+2$$





BOB BOB + BOB

2020 + 2020 = 4040

BE + BE + BE + BE



1234567890



$$n^2_{\infty,alg}(X_{\pm},\infty)\cong BG_{\mp}$$

ERG



$$(10) \rightarrow BG(d) \rightarrow BG(d+2) \rightarrow \cdot \cdot \cdot$$

1990-1991

BC = BD

Q&A



BOEATOP











BOB + 1



